The `Long` class in Java is a wrapper class for the primitive data type `long`. It is part of the `java.lang` package and provides methods to work with long values, such as parsing, converting, and comparing them. The `Long` class also provides constants for representing the maximum and minimum values a `long` can have.

Key Features of Long Wrapper Class

1. **Immutability**: Instances of the `Long` class are immutable.
2. **Constants**: Provides constants like `MAX_VALUE` and `MIN_VALUE`.
3. **Static Methods**: Methods for parsing strings to longs, comparing longs, and converting longs to strings.

Long Class Methods

Here are some commonly used methods in the `Long` class:

1. `byte byteValue()`
2. `int compareTo(Long anotherLong)`
3. `static int compare(long x, long y)`
4. `static Long decode(String nm)`
5. `double doubleValue()`
6. `boolean equals(Object obj)`
7. `float floatValue()`
8. `static Long getLong(String nm)`
9. `static Long getLong(String nm, long val)`
10. `static Long getLong(String nm, Long val)`
11. `int hashCode()`
12. `int intValue()`
13. `long longValue()`
14. `static long parseLong(String s)`
15. `static long parseLong(String s, int radix)`

16. `short shortValue()`
17. `static String toString(long i)`
18. `String toString()`
19. `static Long valueOf(String s)`
20. `static Long valueOf(long l)`

1. `byte byteValue()`

Returns the value of this `Long` as a `byte`.

```
public class LongExample {
    public static void main(String[] args) {
        Long l = 12345L;
        byte value = l.byteValue();
        System.out.println(value);  // Output: 57 (truncated value)
    }
}
```

Output:

```
57
```

2. `int compareTo(Long anotherLong)`

Compares two `Long` objects numerically.

```
public class LongExample {
    public static void main(String[] args) {
        Long l1 = 12345L;
        Long l2 = 54321L;
```

```
        System.out.println(l1.compareTo(l2));  // Output: -1
    }
}
```

Output:

```
-1
```

3. `static int compare(long x, long y)`

Compares two `long` values numerically.

```
public class LongExample {
    public static void main(String[] args) {
        System.out.println(Long.compare(12345L, 54321L));  // Output: -1
    }
}
```

Output:

```
-1
```

4. `static Long decode(String nm)`

Decodes a `String` into a `Long`.

```
public class LongExample {
    public static void main(String[] args) {
```

```
        Long l = Long.decode("12345");
        System.out.println(l);  // Output: 12345
    }
}
```

Output:

```
12345
```

5. `double doubleValue()`

Returns the value of this `Long` as a `double`.

```
public class LongExample {
    public static void main(String[] args) {
        Long l = 12345L;
        double value = l.doubleValue();
        System.out.println(value);  // Output: 12345.0
    }
}
```

Output:

```
12345.0
```

6. `boolean equals(Object obj)`

Compares this object to the specified object.

```java
public class LongExample {
    public static void main(String[] args) {
        Long l1 = 12345L;
        Long l2 = 12345L;
        System.out.println(l1.equals(l2));  // Output: true
    }
}
```

Output:

```
true
```

7. `float floatValue()`

Returns the value of this `Long` as a `float`.

```java
public class LongExample {
    public static void main(String[] args) {
        Long l = 12345L;
        float value = l.floatValue();
        System.out.println(value);  // Output: 12345.0
    }
}
```

Output:

```
12345.0
```

8. `static Long getLong(String nm)`

Returns the `Long` value of the system property with the specified name.

```
public class LongExample {
    public static void main(String[] args) {
        System.setProperty("myLong", "12345");
        Long l = Long.getLong("myLong");
        System.out.println(l);  // Output: 12345
    }
}
```

Output:

```
12345
```

9. `static Long getLong(String nm, long val)`

Returns the `Long` value of the system property with the specified name, or the specified default value if there is no property with that name.

```
public class LongExample {
    public static void main(String[] args) {
        Long l = Long.getLong("nonExistent", 12345L);
        System.out.println(l);  // Output: 12345
    }
}
```

Output:

```
12345
```

10. `static Long getLong(String nm, Long val)`

Returns the `Long` value of the system property with the specified name, or the specified default value if there is no property with that name.

```
public class LongExample {
    public static void main(String[] args) {
        Long l = Long.getLong("nonExistent", Long.valueOf(12345L));
        System.out.println(l);  // Output: 12345
    }
}
```

Output:

```
12345
```

11. `int hashCode()`

Returns a hash code for this `Long` object.

```
public class LongExample {
    public static void main(String[] args) {
        Long l = 12345L;
        System.out.println(l.hashCode());  // Output: a unique hash code
    }
}
```

Output:

```
12345
```

12. `int intValue()`

Returns the value of this `Long` as an `int`.

```
public class LongExample {
    public static void main(String[] args) {
        Long l = 12345L;
        int value = l.intValue();
        System.out.println(value);  // Output: 12345
    }
}
```

Output:

```
12345
```

13. `long longValue()`

Returns the value of this `Long` as a `long`.

```
public class LongExample {
    public static void main(String[] args) {
        Long l = 12345L;
        long value = l.longValue();
        System.out.println(value);  // Output: 12345
```

```
    }
}
```

Output:

```
12345
```

14. `static long parseLong(String s)`

Parses the string argument as a signed decimal long.

```
public class LongExample {
    public static void main(String[] args) {
        long l = Long.parseLong("12345");
        System.out.println(l);  // Output: 12345
    }
}
```

Output:

```
12345
```

15. `static long parseLong(String s, int radix)`

Parses the string argument as a signed long in the radix specified by the second argument.

```
public class LongExample {
    public static void main(String[] args) {
```

```
        long l = Long.parseLong("1101", 2);
        System.out.println(l);  // Output: 13
    }
}
```

Output:

```
13
```

16. `short shortValue()`

Returns the value of this `Long` as a `short`.

```
public class LongExample {
    public static void main(String[] args) {
        Long l = 12345L;
        short value = l.shortValue();
        System.out.println(value);  // Output: 12345
    }
}
```

Output:

```
12345
```

17. `static String toString(long i)`

Returns a `String` object representing the specified `long`.

```
public class LongExample {
    public static void main(String[] args) {
        String s = Long.toString(12345L);
        System.out.println(s);  // Output: "12345"
    }
}
```

Output:

```
12345
```

18. `String toString()`

Returns a `String` object representing this `Long`'s value.

```
public class LongExample {
    public static void main(String[] args) {
        Long l = 12345L;
        System.out.println(l.toString());  // Output: "12345"
    }
}
```

Output:

```
12345
```

19. `static Long valueOf(String s)`

Returns a `Long` object holding the value given by the specified `String`.

```
public class LongExample {
    public static void main(String[] args) {
        Long l = Long.valueOf("12345");
        System.out.println(l);  // Output: 12345
    }
}
```

Output:

```
12345
```

20. `static Long valueOf(long l)`

Returns a `Long` instance representing the specified `long` value.

```
public class LongExample {
    public static void main(String[] args) {
        Long l = Long.valueOf(12345L);
        System.out.println(l);  // Output: 12345
    }
}
```

Output:

```
12345
```